

Highly Available Web Service with Local DNS

Shehab Eldeen Ahmed

July 20, 2026

Contents

1	Task Overview	3
2	Project Structure	3
3	Approach	3
4	Tools Used	3
5	Results	4
5.1	Virtual Machine Availability	4
5.2	DNS Resolution	4
5.3	Primary Web Server	4
5.4	Automatic Failover	5

1 Task Overview

The environment was built as three isolated virtual machines. One machine provides DNS service, while the other two host the same web service. The web servers operate as primary and backup nodes and share a single virtual IP address. Users therefore access one domain name and one service address without needing to know which web server is currently active.

2 Project Structure

The submitted project is organized as follows:

```
linux_task/
├── Vagrantfile
├── provision/
│   ├── dns.sh
│   └── web.sh
├── config/
│   ├── bind/
│   │   ├── db.test.com
│   │   ├── named.conf.local
│   │   └── named.conf.options
│   └── nginx/
│       ├── master.conf
│       ├── backup.conf
│       ├── web1.conf
│       ├── web2.conf
│       ├── web1-index.html
│       └── web2-index.html
├── screenshots/
│   ├── dig_via_the_dns_server.png
│   ├── resolving_the_master.png
│   ├── after_halting_web1.png
│   └── ping.png
└── linux_task_report.tex
```

3 Approach

The environment was created using infrastructure automation so that all three machines could be started and prepared consistently. A dedicated DNS server was used to map the test domain to a virtual IP address. Two web servers were then placed behind that address in a primary–backup arrangement. Health monitoring was used to detect loss of the primary server and transfer the virtual IP address to the backup server automatically.

Validation was carried out in stages: first by checking the virtual machines, then by confirming DNS resolution, accessing the service while the primary was active, and finally stopping the primary server and accessing the same domain again to verify failover.

4 Tools Used

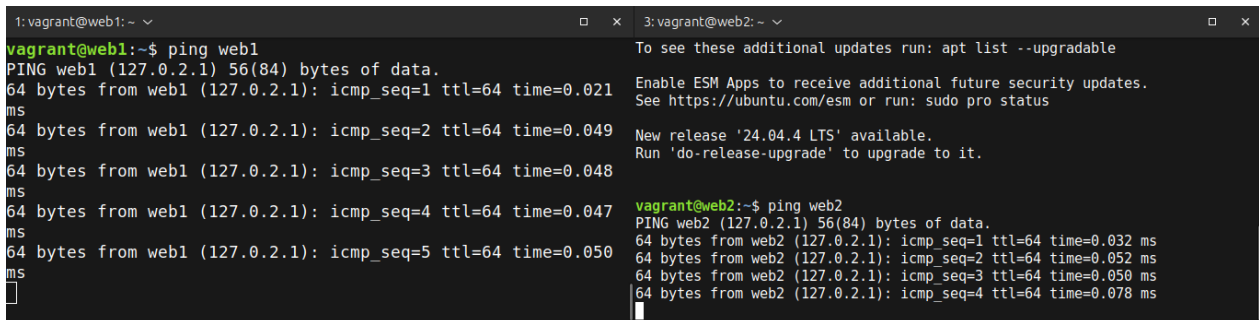
- **Vagrant and VirtualBox** for creating and managing the virtual machines.
- **Ubuntu Linux** as the operating system for all nodes.
- **BIND9** for local DNS resolution.

- **NGINX** for serving the web pages.
- **Keepalived** for health monitoring and virtual IP failover.
- **dig, curl, and ping** for validation and testing.

5 Results

5.1 Virtual Machine Availability

Both web virtual machines were reachable and responsive. This confirmed that the nodes were running successfully before testing the service behavior.



```

1: vagrant@web1: ~
vagrant@web1:~$ ping web1
PING web1 (127.0.2.1) 56(84) bytes of data.
64 bytes from web1 (127.0.2.1): icmp_seq=1 ttl=64 time=0.021 ms
64 bytes from web1 (127.0.2.1): icmp_seq=2 ttl=64 time=0.049 ms
64 bytes from web1 (127.0.2.1): icmp_seq=3 ttl=64 time=0.048 ms
64 bytes from web1 (127.0.2.1): icmp_seq=4 ttl=64 time=0.047 ms
64 bytes from web1 (127.0.2.1): icmp_seq=5 ttl=64 time=0.050 ms

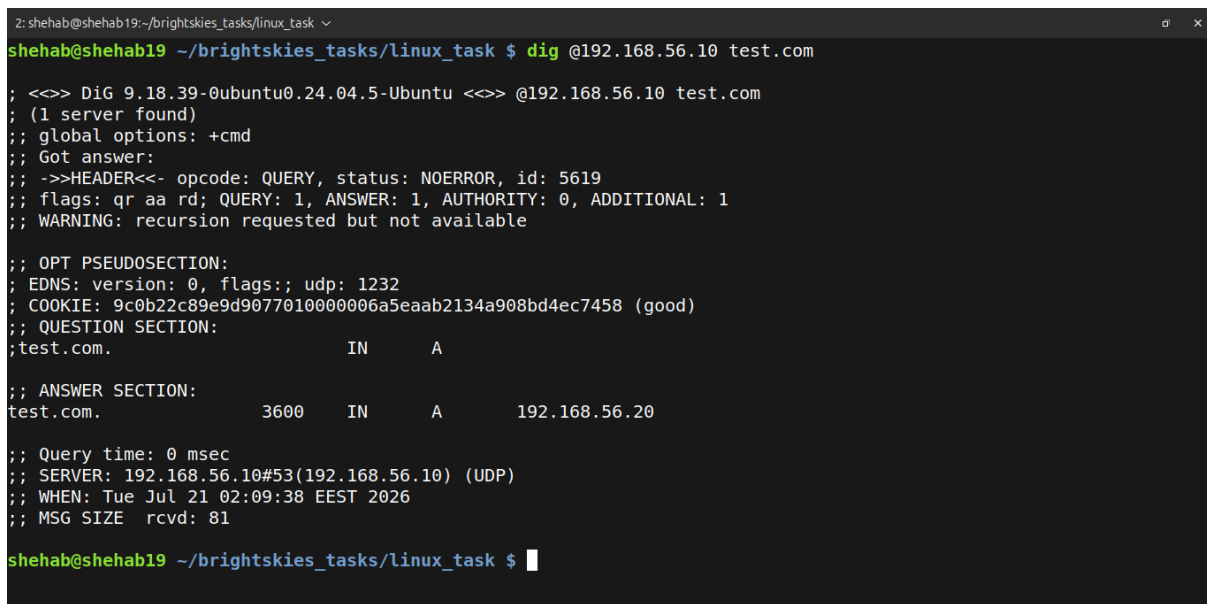
3: vagrant@web2: ~
To see these additional updates run: apt list --upgradable
Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status
New release '24.04.4 LTS' available.
Run 'do-release-upgrade' to upgrade to it.
vagrant@web2:~$ ping web2
PING web2 (127.0.2.1) 56(84) bytes of data.
64 bytes from web2 (127.0.2.1): icmp_seq=1 ttl=64 time=0.032 ms
64 bytes from web2 (127.0.2.1): icmp_seq=2 ttl=64 time=0.052 ms
64 bytes from web2 (127.0.2.1): icmp_seq=3 ttl=64 time=0.050 ms
64 bytes from web2 (127.0.2.1): icmp_seq=4 ttl=64 time=0.078 ms

```

Figure 1: Basic reachability checks on the web-server virtual machines.

5.2 DNS Resolution

The DNS query completed successfully and returned the shared virtual IP address 192.168.56.20 for test.com. This shows that clients can locate the service through its domain name rather than a physical web-server address.



```

2: shehab@shehab19:~/brightskies_tasks/linux_task
shehab@shehab19 ~/~/brightskies_tasks/linux_task $ dig @192.168.56.10 test.com

; <<>> DiG 9.18.39-0ubuntu0.24.04.5-Ubuntu <<>> @192.168.56.10 test.com
; (1 server found)
;; global options: +cmd
;; Got answer:
;; ->HEADER<- opcode: QUERY, status: NOERROR, id: 5619
;; flags: qr aa rd; QUERY: 1, ANSWER: 1, AUTHORITY: 0, ADDITIONAL: 1
;; WARNING: recursion requested but not available

;; OPT PSEUDOSECTION:
; EDNS: version: 0, flags:; udp: 1232
; COOKIE: 9c0b22c89e9d907701000006a5eaab2134a908bd4ec7458 (good)
;; QUESTION SECTION:
;test.com.                IN      A
;; ANSWER SECTION:
test.com.                 3600    IN      A      192.168.56.20

;; Query time: 0 msec
;; SERVER: 192.168.56.10#53(192.168.56.10) (UDP)
;; WHEN: Tue Jul 21 02:09:38 EEST 2026
;; MSG SIZE rcvd: 81

shehab@shehab19 ~/~/brightskies_tasks/linux_task $

```

Figure 2: The local DNS server resolves the domain to the virtual IP address.

5.3 Primary Web Server

With both web servers online, a request to the domain reached Web1, confirming that the primary node initially owned the shared service address and served the expected page.

```
1: vagrant@web1: ~  
vagrant@web1:~$  
3: vagrant@web2: ~  
vagrant@web2:~$  
2: shehab@shehab19:~/brightskies_tasks/linux_task  
shehab@shehab19 ~/brightskies_tasks/linux_task $ curl --resolve test.com:80:$(dig +short @192.168.56.10 test.com) http://test.com  
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <title>Web1</title>  
</head>  
<body>  
  <h1>Master</h1>  
  
  <p>Served by: Web1</p>  
  
  <p>Server IP: 192.168.56.11</p>  
</body>  
</html>  
shehab@shehab19 ~/brightskies_tasks/linux_task $
```

Figure 3: The web service is initially provided by the primary server, Web1.

5.4 Automatic Failover

After Web1 was halted, the same domain and virtual IP address remained available. The response was then served by Web2, demonstrating that the backup node automatically took over the shared address and maintained service availability.

```
1: shehab@shehab19:~/brightskies_tasks/linux_task  
vagrant@web1:~$  
logout  
shehab@shehab19 ~/brightskies_tasks/linux_task $ vagrant halt  
web1  
==> web1: Attempting graceful shutdown of VM...  
shehab@shehab19 ~/brightskies_tasks/linux_task $  
2: shehab@shehab19:~/brightskies_tasks/linux_task  
shehab@shehab19 ~/brightskies_tasks/linux_task $ curl --resolve test.com:80:$(dig +short @192.168.56.10 test.com) http://test.com  
<!DOCTYPE html>  
<html lang="en">  
<head>  
  <meta charset="UTF-8">  
  <title>Web2</title>  
</head>  
<body>  
  <h1>Backup</h1>  
  
  <p>Served by: Web2</p>  
  
  <p>Server IP: 192.168.56.12</p>  
</body>  
</html>  
shehab@shehab19 ~/brightskies_tasks/linux_task $
```

Figure 4: Web2 serves the same domain after the primary server is halted.